Electronic Companion to "Robust Candidate MaxWeight Certificates for Heterogeneous Compute Scheduling: A Scheduleurm Case Study"

EC. Conventional Proof Details

This appendix gives the conventional, paper-readable proof chain behind Theorem 1. It is intentionally redundant with the main text: the purpose is to make clear which part is mathematics, which part is a measured Scheduleurm certificate, and which part is checked by Lean. The Lean artifact is a formal check of the algebraic and recurrence implications; it is not a substitute for empirical certificates of service rows, lower confidence bounds, candidate admission, oracle gaps, or moment bounds.

EC.1. Capacity Slack as a Support-Function Margin

Let $\mu(a) \in \mathbb{R}_+^{|\mathcal{I}|}$ denote the true mean service vector of full action $a \in \mathcal{A}^{\text{full}}$, and let $H_{\mathcal{A}^{\text{full}}}(q) = \max_{a \in \mathcal{A}^{\text{full}}} q^\top \mu(a)$ be the full-action support function for nonnegative backlog weights $q \in \mathbb{R}_+^{|\mathcal{I}|}$. A stationary randomized policy over full actions is a probability vector $x \in \text{Mix}(\mathcal{A}^{\text{full}})$. If it has coordinate slack $\delta > 0$ against the arrival mean λ, then

$$\lambda_i + \delta \le \sum_{a \in \mathcal{A}^{\text{full}}} x_a \mu_i(a), \qquad i \in \mathcal{I}. \tag{EC.1}$$

Multiplying each coordinate by $q_i \ge 0$ and summing gives

$$q^\top \lambda + \delta \|q\|_1 \le \sum_{a \in \mathcal{A}^{\text{full}}} x_a q^\top \mu(a). \tag{EC.2}$$

Because x is a convex combination, the right-hand side is at most the maximum full-action support:

$$\sum_{a \in \mathcal{A}^{\text{full}}} x_a q^\top \mu(a) \le H_{\mathcal{A}^{\text{full}}}(q). \tag{EC.3}$$

Hence full-action capacity slack implies

$$q^\top \lambda + \delta \|q\|_1 \le H_{\mathcal{A}^{\text{full}}}(q). \tag{EC.4}$$

This is the only place where the capacity-region hypothesis enters the drift proof. It converts an interior capacity statement into a MaxWeight support margin that can be compared to candidate actions. The paper uses the support-function statement as Assumption 1; the operational necessity direction is separate and requires a load-certified stochastic model and a conservation law.

EC.2. Candidate Fabric Cover and Support Loss

The scheduler cannot enumerate every full configuration. It evaluates an admitted candidate family $\mathcal{A}_t^{\mathrm{cand}} \subseteq \mathcal{A}^{\mathrm{full}}$, possibly depending on the observed state t. The finite-candidate theorem needs a support loss certificate: for every $q \geq 0$,

$$H_{\mathcal{A}^{\mathrm{full}}}(q) \leq H_{\mathcal{A}_t^{\mathrm{cand}}}(q) + \epsilon_{\mathrm{cand}} \|q\|_1 . \tag{EC.5}$$

When the loss is obtained from a calibrated fabric metric, let $\Phi(a)$ be the finite feature representation of action a, let d_Φ be the feature metric, and suppose each full action a has a projected candidate $\pi_t(a) \in \mathcal{A}_t^{\mathrm{cand}}$ satisfying

$$d_\Phi(a, \pi_t(a)) \leq \rho. \tag{EC.6}$$

If service is Lipschitz in this representation,

$$\|\mu(a) - \mu(\pi_t(a))\|_\infty \leq L\, d_\Phi(a, \pi_t(a)), \tag{EC.7}$$

then for any maximizer $a^\star \in \arg\max_{a \in \mathcal{A}^{\mathrm{full}}} q^\top \mu(a)$,

$$\begin{aligned} H_{\mathcal{A}^{\mathrm{full}}}(q) &= q^\top \mu(a^\star) \\ &\leq q^\top \mu(\pi_t(a^\star)) + \|q\|_1 L\rho \\ &\leq H_{\mathcal{A}_t^{\mathrm{cand}}}(q) + L\rho \|q\|_1 . \end{aligned} \tag{EC.8}$$

Thus $\epsilon_{\mathrm{cand}} = L\rho$ in the fabric-cover case. In measured finite slices where the candidate support is certified directly, the artifact uses the observed support gap itself as ϵ_{cand}; when the candidate set is exact on the measured slice, this term is zero. This is why the proof does not shrink the action space informally: every restriction must be paid for by the explicit support-function loss in (EC.5).

Proposition 1 supplies a constructive version of this step. It forms candidates as representatives of active feature cells, bounds the candidate cardinality by the number of cells, and derives the same $L\rho$ support loss. Thus the theorem can consume either a directly audited support gap or a finite-cell construction certificate; only the latter supports a claim of nontrivial action-space compression.

EC.3. Lower-Service Rows, Penalties, and Approximate Oracles

The robust scheduler optimizes conservative service, not the unknown true mean service. For each admitted candidate $a \in \mathcal{A}_t^{\mathrm{cand}}$, the artifact provides a lower-service vector $\mu_t^{\mathrm{L}}(a)$. The lower-service support loss is

$$H_{\mathcal{A}_t^{\mathrm{cand}}}(q) \leq \max_{a \in \mathcal{A}_t^{\mathrm{cand}}} q^\top \mu_t^{\mathrm{L}}(a) + \epsilon_{\mathrm{est}} \|q\|_1 . \tag{EC.9}$$

The scheduling score is

$$\Psi_t(a) = Q^\top \mu_t^{\mathrm{L}}(a) - K_t(a) - G_t(a), \tag{EC.10}$$

where $K_t(a)$ and $G_t(a)$ are bounded switching, risk, or guard penalties. Equation (EC.9) is the lower-confidence-bound bridge from measured service rows to the drift proof. The coordinate-scaled sufficient condition for this bridge is given in Proposition 2: on the simultaneous confidence event it charges $2\epsilon_{\mathrm{LCB}} \sum_i s_i q_i$, or $2\epsilon_{\mathrm{LCB}} C_s \|q\|_1$ under a uniform scale bound. This prevents a high-throughput workload from defining the absolute error unit for all other classes. The selected action a_t need not be the exact maximizer. It satisfies

$$\max_{a \in \mathcal{A}_t^{\mathrm{cand}}} \Psi_t(a) - \Psi_t(a_t) \le \alpha_0 + \alpha_1 \|Q\|_1, \tag{EC.11}$$

and all admitted candidate penalties obey

$$0 \le K_t(a) + G_t(a) \le P_0 + \beta \|Q\|_1, \qquad a \in \mathcal{A}_t^{\mathrm{cand}}. \tag{EC.12}$$

Combining the capacity support bound (EC.4), the two support-loss bounds (EC.5) and (EC.9), the oracle gap (EC.11), and the penalty bound (EC.12) yields the selected lower-service margin. First,

$$\max_{a \in \mathcal{A}_t^{\mathrm{cand}}} Q^\top \mu_t^{\mathrm{L}}(a) \ge Q^\top \lambda + (\delta - \epsilon_{\mathrm{cand}} - \epsilon_{\mathrm{est}}) \|Q\|_1. \tag{EC.13}$$

Second, by the oracle gap and the all-candidate penalty bound,

$$\begin{aligned} Q^\top \mu_t^{\mathrm{L}}(a_t) &\ge \max_{a \in \mathcal{A}_t^{\mathrm{cand}}} Q^\top \mu_t^{\mathrm{L}}(a) \\ &\quad - (P_0 + \alpha_0) - (\beta + \alpha_1) \|Q\|_1 \\ &\ge Q^\top \lambda + \eta \|Q\|_1 - (P_0 + \alpha_0), \end{aligned} \tag{EC.14}$$

where

$$\eta = \delta - (\epsilon_{\mathrm{cand}} + \epsilon_{\mathrm{est}} + \beta + \alpha_1). \tag{EC.15}$$

The theorem therefore has a simple accounting rule: candidate loss, lower-service estimation loss, queue-scaled penalty, and queue-scaled oracle loss all consume the same capacity slack. A stability certificate requires $\eta > 0$. Additive terms P_0 and α_0 do not change the slope of the drift; they enlarge the finite set outside which the drift is negative.

EC.4. Quadratic Drift and the Foster Certificate

The queue update is

$$Q_i(t+1) = \left[Q_i(t) - S_i(t)\right]^+ + A_i(t), \tag{EC.16}$$

where $A_i(t)$ is arrival and $S_i(t)$ is potential service for class i. Completed departure is $D_i(t) = \min\{Q_i(t), S_i(t)\}$. The displayed update remains valid because unused potential service is removed by the positive part. For $V(Q) = \frac{1}{2}\sum_i Q_i^2$, the standard one-step inequality gives

$$\begin{aligned}\Delta V(t) \le \frac{1}{2}\sum_i \left(A_i(t)^2 + S_i(t)^2\right) \\ + Q(t)^\top A(t) - Q(t)^\top S(t).\end{aligned} \tag{EC.17}$$

Assumption 4 supplies the conditional second-order bound

$$\mathbb{E}\left[\frac{1}{2}\sum_i \left(A_i(t)^2 + S_i(t)^2\right) \middle| \mathcal{F}_t\right] \le B. \tag{EC.18}$$

It also ties the stochastic process to the theorem-facing service rows:

$$\mathbb{E}[A(t) \mid \mathcal{F}_t] \le \lambda, \tag{EC.19}$$

$$\mathbb{E}[S(t) \mid \mathcal{F}_t] \ge \mu_t^{\mathrm{L}}(a_t) \tag{EC.20}$$

coordinatewise. Taking conditional expectation in (EC.17) and using (EC.14),

$$\begin{aligned}\mathbb{E}[\Delta V(t) \mid \mathcal{F}_t] &\le B + Q(t)^\top \lambda - Q(t)^\top \mu_t^{\mathrm{L}}(a_t) \\ &\le B + P_0 + \alpha_0 - \eta \left\|Q(t)\right\|_1 .\end{aligned} \tag{EC.21}$$

If N and $\gamma > 0$ satisfy

$$B + P_0 + \alpha_0 + \gamma \le \eta N, \tag{EC.22}$$

then whenever $\|Q(t)\|_1 > N$, (EC.21) implies

$$\mathbb{E}[\Delta V(t) \mid \mathcal{F}_t] \le -\gamma. \tag{EC.23}$$

This is the finite-set drift certificate used in Theorem 1. To translate the certificate into positive recurrence for a Markov chain, the paper also assumes a countable augmented state space, a finite nonempty backlog sublevel set, and a closed communicating class or equivalent local-return condition. Without those recurrence-side conditions the theorem should be read as a negative-drift certificate, not as an unconditional positive-recurrence statement for every possible implementation.

EC.5. Statewise Feasible Families and Operational Boundaries

The statewise corollary is a conditioning argument rather than a new drift identity. Let $\chi(t)$ record scheduler-visible information such as alive nodes, pinned tasks, queued jobs, and measured admission status. If the full family, candidate family, true and lower-service maps, feature map, penalty bounds, oracle bounds, and moment bounds are indexed by $X(t) = (Q(t), \chi(t))$ and satisfy the same constants for every admitted state, then the proof in EC.1–EC.4 holds after conditioning on $\mathcal{F}_t$. Uniform constants preserve both the drift and finite-horizon backlog bounds. Positive recurrence additionally requires the augmented-state conditions in Assumption 5; it does not follow merely by averaging over $\chi(t)$.

The operational capacity statement is kept separate. Sufficiency follows from the positive-slack drift chain above once a stochastic model certifies the arrival mean, service accounting, lower-service domination, moment bounds, and local recurrence conditions. Necessity is not obtained by defining a model to "stabilize" a load; it requires the model to encode the load and obey a conservation law showing that long-run departures cannot exceed the capacity base. This separation is essential for the paper's claim boundary: Lean checks the implication from certified assumptions to drift and recurrence, while the experiments and admission checks certify when Scheduleurm rows satisfy those assumptions.

Call an integer-queue model *load certified for* λ when it has an explicit finite-support arrival/service realization with the same transition kernel, conditional mean arrival equal to λ in every state, and conditional mean service equal to the service accounting used by the model. Call the load operationally stabilizable when such a model is positive recurrent through a finite backlog sublevel set. This binding prevents a stable but unrelated transition kernel from serving as a witness for an arbitrary load.

For necessity, the model-specific conservation law is the following statement. Every operationally stable load-certified model admits a stationary occupation mix $x^{\mathrm{occ}} \in \mathrm{Mix}(\mathcal{A}^{\mathrm{full}})$ satisfying

$$\lambda_i \le \sum_{a \in \mathcal{A}^{\mathrm{full}}} x_a^{\mathrm{occ}} \mu_i(a), \qquad i \in \mathcal{I}. \tag{EC.24}$$

The occupation mix is a model certificate, not an automatic consequence of the geometric definition of $\Lambda(\mathcal{A}^{\mathrm{full}})$.

PROPOSITION EC.1 **(Load-certified operational capacity sandwich)**. *Suppose the full action family and service map satisfy both of the following model-specific obligations.*

(i) For every λ with positive full-action slack, there is a load-certified integer-queue model satisfying the hypotheses of Theorem 1 and Assumption 5.

(ii) Every operationally stable load-certified model satisfies the conservation law (EC.24).

Then positive-slack membership in $\Lambda(\mathcal{A}^{\mathrm{full}})$ is sufficient for operational stabilizability, and operational stabilizability is necessary only up to the zero-slack closure:

$$\begin{aligned} &\lambda \text{ operationally stabilizable} \\ &\Longrightarrow \lambda \in \Lambda(\mathcal{A}^{\mathrm{full}}). \end{aligned} \tag{EC.25}$$

Thus the result is a positive-slack/zero-slack sandwich, not an unsupported equivalence at a fixed positive slack.

Proof. For sufficiency, obligation (i) supplies a load-certified model whose arrival mean is the same λ appearing in the capacity premise. The positive residual margin in Theorem 1 gives negative drift outside a finite backlog sublevel set, and Assumption 5 turns that drift certificate into positive recurrence of the relevant closed class. Hence the same load λ is operationally stabilizable.

For necessity, take an operationally stable load-certified model. Obligation (ii) supplies $x^{\mathrm{occ}} \in \mathrm{Mix}(\mathcal{A}^{\mathrm{full}})$ satisfying (EC.24). Define $v_i = \sum_{a \in \mathcal{A}^{\mathrm{full}}} x_a^{\mathrm{occ}} \mu_i(a)$. By the definition of the stationary mixture, $v \in C(\mathcal{A}^{\mathrm{full}})$, and the conservation law gives $\lambda \le v$ coordinatewise. The definition of the downward-closed region therefore gives $\lambda \in \Lambda(\mathcal{A}^{\mathrm{full}})$, proving (EC.25). The two directions use different model-specific premises, which is why no positive-slack "if and only if" is claimed.

EC.6. Variable-Duration Configuration Trajectories

The slotted theorem does not silently cover migration, machine reconfiguration, or port moves whose durations differ. We therefore state the separate frame result used by the trajectory interface. Let n index decision frames, let $\mathcal{F}_n^{\mathrm{F}}$ be the information available at the start of frame n, and let the $\mathcal{F}_n^{\mathrm{F}}$-measurable duration satisfy

$$0 < \tau_{\min} \le \tau_n \le \tau_{\max} < \infty. \tag{EC.26}$$

The frame-boundary queue obeys

$$Q^{\mathrm{F}}(n+1) = \left[Q^{\mathrm{F}}(n) - S^{\mathrm{F}}(n)\right]^+ + A^{\mathrm{F}}(n), \tag{EC.27}$$

where $A^{\mathrm{F}}(n)$ and $S^{\mathrm{F}}(n)$ are cumulative arrivals and potential service during the frame. Define

$$\overline{S}^{\mathrm{F}}(n) = \mathbb{E}\left[S^{\mathrm{F}}(n) \mid \mathcal{F}_n^{\mathrm{F}}\right]. \tag{EC.28}$$

Suppose the conditional arrival and second-order bounds are

$$\mathbb{E}\left[A_i^{\mathrm{F}}(n) \mid \mathcal{F}_n^{\mathrm{F}}\right] \le \tau_n \lambda_i, \tag{EC.29}$$

$$\mathbb{E}\left[\frac{1}{2}\sum_i \left\{(A_i^{\mathrm{F}}(n))^2 + (S_i^{\mathrm{F}}(n))^2\right\} \middle| \mathcal{F}_n^{\mathrm{F}}\right] \le B_{\mathrm{F}}. \tag{EC.30}$$

Let ϵ_{F} combine candidate-support and conservative-service losses in rate units. The cumulative approximate-oracle and penalty obligations are

$$\begin{aligned}\tau_n H_{\mathcal{A}^{\mathrm{full}}}\left(Q^{\mathrm{F}}(n)\right) \le \left(Q^{\mathrm{F}}(n)\right)^{\top} \overline{S}^{\mathrm{F}}(n) \\ + \tau_n(\epsilon_{\mathrm{F}} + \alpha_1)\left\|Q^{\mathrm{F}}(n)\right\|_1 \\ + \alpha_0 + P_n^{\mathrm{F}}.\end{aligned} \tag{EC.31}$$

$$P_n^{\mathrm{F}} \le P_0 + \tau_n \beta \left\|Q^{\mathrm{F}}(n)\right\|_1. \tag{EC.32}$$

All quantities in (EC.31)–(EC.32) are in cumulative work units over the same frame; in particular, migration lost work cannot be inserted as an unmatched wall-clock scalar. For the controlled migration rows, let $W_n^{\mathrm{rem}} > 0$ be one task's remaining work, let $\underline{\mu}_n^{\mathrm{old}}$ and $\underline{\mu}_n^{\mathrm{new}}$ be its per-task lower-service rates in the measured source and destination states, and let T_n^{mig} be the measured checkpoint, synchronization, staging, warmup, lost-work, and time-equivalent risk overhead. The migration trajectory is represented by the duration-normalized lower service

$$\underline{\mu}_n^{\mathrm{mig,eff}} = \frac{W_n^{\mathrm{rem}}}{T_n^{\mathrm{mig}} + W_n^{\mathrm{rem}}/\underline{\mu}_n^{\mathrm{new}}}. \tag{EC.33}$$

It enters the candidate family only when

$$\frac{W_n^{\mathrm{rem}}}{\underline{\mu}_n^{\mathrm{old}}} - \frac{W_n^{\mathrm{rem}}}{\underline{\mu}_n^{\mathrm{new}}} > T_n^{\mathrm{mig}}, \tag{EC.34}$$

equivalently $\underline{\mu}_n^{\mathrm{mig,eff}} > \underline{\mu}_n^{\mathrm{old}}$. Thus wall-clock overhead first changes the trajectory duration; it is not subtracted directly from a service-rate score.

For the detailed proof of Proposition 3, suppose (EC.26)–(EC.32) hold and the full family has support slack δ. Define

$$\eta_{\mathrm{F}} = \delta - \epsilon_{\mathrm{F}} - \alpha_1 - \beta. \tag{EC.35}$$

Then

$$\begin{aligned}
&\mathbb{E}\left[V\left(Q^{\mathrm{F}}(n+1)\right) - V\left(Q^{\mathrm{F}}(n)\right)\middle|\mathcal{F}_n^{\mathrm{F}}\right] \\
&\qquad \le B_{\mathrm{F}} + \alpha_0 + P_0 - \tau_n \eta_{\mathrm{F}} \left\|Q^{\mathrm{F}}(n)\right\|_1 \\
&\qquad \le B_{\mathrm{F}} + \alpha_0 + P_0 - \tau_{\min} \eta_{\mathrm{F}} \left\|Q^{\mathrm{F}}(n)\right\|_1,
\end{aligned} \tag{EC.36}$$

provided $\eta_{\mathrm{F}} > 0$. If the frame-boundary augmented state is countable, its backlog sublevel sets are finite and nonempty, and the usual local-return condition holds, then any $N \in \mathbb{N}$ and $\gamma > 0$ satisfying

$$B_{\mathrm{F}} + \alpha_0 + P_0 + \gamma \le \tau_{\min} \eta_{\mathrm{F}} N \tag{EC.37}$$

give a finite-set Foster recurrence certificate for the embedded frame-boundary chain. Moreover, elapsed physical time through frame n is at most $n\tau_{\max}$.

Detailed proof of Proposition 3. The quadratic queue inequality applied to (EC.27), followed by conditional expectation, gives

$$\begin{aligned}
\mathbb{E}[\Delta_{\mathrm{F}} V(n) \mid \mathcal{F}_n^{\mathrm{F}}] \le B_{\mathrm{F}} + \tau_n \left(Q^{\mathrm{F}}(n)\right)^{\top} \lambda \\
- \left(Q^{\mathrm{F}}(n)\right)^{\top} \overline{S}^{\mathrm{F}}(n),
\end{aligned} \tag{EC.38}$$

where $\Delta_{\mathrm{F}} V(n)$ denotes the frame-to-frame Lyapunov increment. Full-action support slack implies

$$\left(Q^{\mathrm{F}}(n)\right)^{\top} \lambda + \delta \left\|Q^{\mathrm{F}}(n)\right\|_1 \le H_{\mathcal{A}^{\mathrm{full}}}\left(Q^{\mathrm{F}}(n)\right). \tag{EC.39}$$

Multiplying (EC.39) by τ_n, substituting (EC.31) and (EC.32) into (EC.38), and collecting coefficients yields the first inequality in (EC.36). Positivity of η_{F} and $\tau_n \ge \tau_{\min}$ gives the second. Equation (EC.37) makes the conditional drift at most $-\gamma$ outside the finite sublevel set, so the same local-return argument as in EC.4 applies to the embedded chain. Finally, $\sum_{k=0}^{n-1} \tau_k \le n\tau_{\max}$ proves the calendar-time bound.

If arrivals and service are rate-scaled throughout a frame and their unit-time second-order term is at most B, then the formal artifact also checks the constructive choice $B_{\mathrm{F}} \le \tau_{\max}^2 B$. Setting $\tau_n = \tau_{\min} = \tau_{\max} = 1$, $B_{\mathrm{F}} = B$, and $\epsilon_{\mathrm{F}} = \epsilon_{\mathrm{cand}} + \epsilon_{\mathrm{est}}$ recovers the slotted drift theorem. Thus the result strengthens rather than relabels the migration boundary: a measured migration row must still certify cumulative net service, duration, checkpoint/synchronization/resume loss, and risk before it can enter a stochastic frame model.

EC.7. Conditional Hidden-Regime and Active-Bucket Extensions

These results are formalized extensions, not premises silently used by Theorem 1. They state exactly what a future sampler or regime detector must certify before a probabilistic learning or switching claim can be composed with the stability theorem.

Let $\mathcal{T}$ be a finite set of decision slots partitioned by a regime segment map $\sigma : \mathcal{T} \to \mathcal{Z}_{\mathrm{seg}}$, and let $\mathcal{M} \subseteq \mathcal{T}$ be the marked switching or detection-delay slots. Write $\mathfrak{D}_t$ for the one-step drift quantity, ℓ_t^{est} for its estimation-loss term, and B_{d} for the slotwise additive drift bound. Let $\eta_0 \geq 0$ be the base drift margin. Suppose

$$\mathfrak{D}_t \leq B_{\mathrm{d}} - \eta_0 \|Q(t)\|_1 + \ell_t^{\mathrm{est}}, \qquad t \in \mathcal{T}, \tag{EC.40}$$

and each segment $z \in \mathcal{Z}_{\mathrm{seg}}$ satisfies the marked-backlog budget

$$\sum_{\substack{t \in \mathcal{M} \\ \sigma(t)=z}} \|Q(t)\|_1 \leq \theta_{\mathrm{d}} \sum_{\substack{t \in \mathcal{T} \\ \sigma(t)=z}} \|Q(t)\|_1 + \zeta_z, \tag{EC.41}$$

where $\theta_{\mathrm{d}} \geq 0$ is the dwell/detection coefficient and $\zeta_z \geq 0$ is the fixed segment budget. Let $u_{\mathrm{sw}} \geq 0$ be the coefficient charged to marked backlog and let C_{sw} be an additive switching cost with $C_{\mathrm{sw}} \geq 0$.

For the learning extension, let $\mathcal{B}_{\mathrm{act}}$ be the finite set of active fabric or co-location buckets. At slot t, let ε_t be the nonnegative active-bucket confidence radius and let $R_{\mathcal{T}}$ be the cumulative true-support minus lower-confidence-support loss. Suppose the simultaneous confidence path gives

$$R_{\mathcal{T}} \leq 2 \sum_{t \in \mathcal{T}} \varepsilon_t \|Q(t)\|_1, \tag{EC.42}$$

and the active-bucket concentration certificate gives

$$\sum_{t \in \mathcal{T}} \varepsilon_t \leq C_{\mathrm{act}} \sqrt{|\mathcal{B}_{\mathrm{act}}| \nu_{\mathcal{T}}}, \tag{EC.43}$$

where $C_{\mathrm{act}} \geq 0$ is a concentration constant and $\nu_{\mathcal{T}} \geq 0$ is the certified horizon mass.

Proposition EC.2 **(Conditional switching and active-bucket bounds)**. *Under the two preceding assumptions,*

$$\begin{aligned} &\sum_{t \in \mathcal{T}} \mathfrak{D}_t + u_{\mathrm{sw}} \sum_{t \in \mathcal{M}} \|Q(t)\|_1 + C_{\mathrm{sw}} \\ &\quad \leq |\mathcal{T}| B_{\mathrm{d}} - (\eta_0 - u_{\mathrm{sw}} \theta_{\mathrm{d}}) \sum_{t \in \mathcal{T}} \|Q(t)\|_1 + \sum_{t \in \mathcal{T}} \ell_t^{\mathrm{est}} \\ &\qquad + u_{\mathrm{sw}} \sum_{z \in \mathcal{Z}_{\mathrm{seg}}} \zeta_z + C_{\mathrm{sw}}. \end{aligned} \tag{EC.44}$$

Under (EC.42)–(EC.43)*,* $Q_{\max} \geq 0$*, and* $\|Q(t)\|_1 \leq Q_{\max}$ *for all* t*,*

$$R_{\mathcal{T}} \leq 2Q_{\max}C_{\mathrm{act}}\sqrt{|\mathcal{B}_{\mathrm{act}}|\nu_{\mathcal{T}}}. \tag{EC.45}$$

If each active-bucket event E_b *fails with probability at most* $p_b \in [0,1]$*, then the simultaneous event fails with probability at most* $\sum_{b\in\mathcal{B}_{\mathrm{act}}} p_b$*. More generally, if a measurable certificate event has probability at least* $p \in [0,1]$ *and implies a deterministic Foster certificate, then that Foster certificate holds with probability at least* p*.*

Proof. Sum (EC.40) over $\mathcal{T}$, partition the marked backlog sum by $\sigma(t)$, multiply (EC.41) by u_{sw}, and add the resulting inequality. Collecting the total-backlog coefficient gives $\eta_0 - u_{\mathrm{sw}}\theta_{\mathrm{d}}$, which proves (EC.44). For the learning bound, substitute $\|Q(t)\|_1 \leq Q_{\max}$ into (EC.42) and then apply (EC.43). The probability statement follows because the complement of $\bigcap_{b\in\mathcal{B}_{\mathrm{act}}} E_b$ is the union of the bucket-level complements; the finite union bound gives the stated failure probability. The last statement is event inclusion followed by monotonicity of probability.

The bounds are deliberately conditional. The Lean implications do not prove that Scheduleurm's adaptive sampler attains (EC.43), nor that a deployed detector attains (EC.41); those are separate stochastic-model and live-validation obligations.

EC.8. Lean Crosswalk and Artifact Assumptions

Tables EC.1, EC.2, and EC.3 separate the formal proof contract from the empirical Scheduleurm certificates. The first two tables give exact, searchable Lean identifiers for every paper-facing formal statement. The supplementary theorem crosswalk additionally indexes lower-level helper lemmas; those helpers are part of the complete checked source but are not presented as independent paper claims.

The stepwise EC.1–EC.4 chain is itself searchable rather than hidden inside a single wrapper theorem. Its principal Lean checkpoints are: *EC.1:* `capacity_slack_implies_support_slack`.
EC.2: `main_candidate_restricted_capacity_approximation` and
`main_candidate_restricted_capacity_approximation_from_calibration`.
EC.3: `main_robust_candidate_maxweight_drift_approx_oracle`.
EC.4: `expectedLyapunovDrift_eq` and
`main_concrete_finite_support_stochastic_stability_from_coordinate_moments`.

The complete artifact crosswalk records the subordinate algebraic and recurrence lemmas used between these checkpoints.

The submitted Lean file is the complete 6,680-line checked source, not an excerpt generated from the two tables. The tables map every paper-facing formal statement to exact declarations; the accompanying theorem crosswalk also maps every split-source module back to the main text or EC layer that uses it. The EC therefore represents the full formal development semantically and searchably without typesetting all implementation lemmas as additional paper claims.

Table EC.1 Core paper statements and searchable Lean theorems.

Paper statement	Searchable Lean theorem(s)	Formal coverage
Proposition 1	`finite_feature_cell_candidate_certificate`	Containment, active-cell cardinality, and $L\rho$ support loss.
Proposition 2	`main_diagonal_scaled_lcb_support_loss`; `main_diagonal_scaled_lcb_support_loss_l1`	Coordinate scaling and the induced lower-service support loss.
Theorem 1, exact-oracle case	`main_theorem_robust_candidate_maxweight_stability_with_second_moment_bound`	Candidate and estimation loss, bounded moment, drift, and recurrence.
Theorem 1, approximate-oracle case	`main_theorem_robust_candidate_maxweight_stability_with_second_moment_bound_approx_oracle`	Additive and queue-scaled oracle loss in the same slack budget.
Theorem 1, calibrated fabric specialization	`main_theorem_robust_candidate_maxweight_stability_from_calibrated_fabric_with_second_moment_bound_approx_oracle`	Projection, feature sensitivity, nonzero $L\rho$, and approximate oracle.
Corollary 2	`main_statewise_calibrated_fabric_robust_candidate_stability_with_second_moment_bound_approx_oracle`	Queue-indexed feasible families with uniform constants.
Corollary 1	`foster_telescoping_l1_bound`	Finite-horizon and asymptotic expected backlog bounds.

Every identifier is searchable in `ScheduleurmUpload.lean`. The main-text statements are conventional versions of these formal implications; the stated augmented-filtration boundary for Corollary 2 remains explicit.

Table EC.2 Additional paper statements and searchable Lean theorems.

Paper statement	Searchable Lean theorem(s)	Formal coverage
Proposition 3 and EC.6	`frameApproximateOracle_iff_durationNormalized`; `frame_approximate_maxWeight_lyapunov_drift_uniform`; `frame_nat_model_positive_recurrent_via_finite_set`; `frame_lyapunov_drift_duration_one`; `elapsedFrameTime_le`	Cumulative units, uniform frame drift, embedded recurrence, slotted reduction, and elapsed time.
Proposition 4	`support_gap_mono_under_candidate_expansion`; `candidate_capacity_slack_loss_under_expansion`	Support-loss and capacity-margin monotonicity under finite action-family expansion.
Proposition EC.1	`ModelEncodesLoad`; `OperationallyStabilizesIntegerLoad`; `main_operational_conservation_law_necessity`; `main_operational_capacity_sandwich`	Load binding, positive-slack sufficiency, and zero-slack conservation-law necessity.
Proposition EC.2, switching part	`main_hidden_regime_dwell_switching_drift`	Explicit dwell/detection budget and switching charge; extension only.
Proposition EC.2, learning part	`main_active_bucket_lcb_learning_regret`; `main_active_bucket_lcb_learning_regret_high_probability`; `main_active_bucket_local_failure_union_bound`; `main_high_probability_stability_from_certificate_event`	Active-bucket regret, finite union bound, and event-to-certificate lifting; extension only.

These rows close the reverse map from each paper-level formalization claim to the submitted Lean source. They do not promote the conditional extension rows to main empirical claims.

Table EC.3 Formal hypotheses and Scheduleurm certificates.

Object	Formal role	Scheduleurm certificate or boundary
λ, δ	Arrival mean and full-action capacity slack.	Online arrival replay, load sweep, and finite lower-service capacity checks.
ϵ_{cand} or $L\rho$	Candidate-set or fabric-cover support loss.	Measured finite-slice support gaps, declared bucket admission, and fabric-cover calibration rows when available.
μ_t^{L}, ϵ_{est}	Conservative lower-service rows and estimation loss.	Holdout lower confidence bound checks, measured service cache, and admission/probe status.
P_0, β	Bounded and queue-scaled penalty consumption.	Penalty-unit records, guard/tie-break checks, and bounded switching configuration.
α_0, α_1	Additive and queue-scaled approximate oracle loss.	Oracle-gap check comparing selected action against theorem-facing candidate families.
B, N, γ	Second-order moment bound and finite-set drift threshold.	Workload bounds, controlled trace statistics, and recurrence-side finite-state admission conditions.
$\tau_{\min}, \tau_{\max}, B_{\mathrm{F}}$	Frame-duration bounds and cumulative second-order term.	A trajectory row must use natural elapsed duration and cumulative arrivals/service; unit-rate scaling yields the audited bound $B_{\mathrm{F}} \le \tau_{\max}^2 B$ when applicable.
$\overline{S}^{\mathrm{F}}(n)$, P_n^{F}	Conditional cumulative service and frame-level migration/reconfiguration loss.	Checkpoint flush, synchronization, staging, resume warmup, lost work, and failure risk must be converted to the same cumulative service/penalty units; deterministic domain holdouts alone do not certify this stochastic bridge.
Statewise $\chi(t)$	Scheduler-visible feasible-family index.	Alive node, pinned-job, queued-job, service-cache, and admission-state trace fields.
Operational model certificate	Links stochastic arrivals and services to the queueing theorem.	Load-certified model contract plus conservation-law boundary; not inferred from Lean alone.

The table is deliberately conservative: Lean verifies the implications once these hypotheses hold, while the Scheduleurm artifact determines which measured rows satisfy them.

Appendix: Supplementary Evidence Index

The full machine-readable status dashboard is provided in the supplementary artifact package. Tables EC.4 and EC.5 give a compact index of what is included and what each evidence family is allowed to support. They summarize the detailed supplementary status dashboard in manuscript-facing evidence categories; the row-level dashboard remains in the supplement, so the compression changes presentation rather than the underlying evidence.

Table EC.4 Supplementary evidence index: theorem and Scheduleurm certificates.

Gate family	Retained dashboard evidence	Boundary kept in manuscript
Formal proof package	Consolidated Lean upload, toolchain, build log, theorem crosswalk, no-`sorry/admit/axiom` proof search, and checksums.	Formalizes theorem implications only after the stated service, moment, slack, and recurrence assumptions are certified.
Declared finite-domain cover	q00/q01/q10/q11 service-cache summaries, admission boundaries, measured support gaps, all-state safety cover, and probe/defer status for unknown states.	Supports declared finite measured domains; it does not assert positive service for arbitrary future states.
Lower-service and slack certificates	Holdout lower confidence bound (LCB) rows, selected-profile stochastic LCB certificate, row-level lower-service slack, $L\rho$ accounting, and finite second-moment thresholds.	Certifies lower-service capacity for admitted rows; mean-service and broad fabric-cover claims require additional calibration.
Replay and online stress	Explicit task-list replay, Poisson and bursty load sweeps, ablations, guarded-policy comparisons, and slack accounting tables.	Compares policies on the same measured service cache; it is not a proof that a static replay trace is a stationary arrival process.
Controlled launch/completion	Six-task and 32-task theorem-admitted launches, completion records, candidate rows, and $\alpha_0 = \alpha_1 = 0$ oracle-gap checks.	Supports controlled hook execution; it is separate from production-wide organic dispatch claims.
Production-history bridge	Strict completed-active production classification, strict launched/completed counts, read-only shadow traces, service-map oracle bridge, and exclusion of attempted-only or external rows.	Supports admitted history evidence; raw telemetry and future unknown jobs are outside the theorem population.
Live-oracle and global-dispatch readiness	Optional global-batch soft-hint hook, live-oracle trace fields, active-production shadow rows, and production readiness bridge.	Shows hook readiness and read-only conversion to theorem rows; it does not make the global batch policy the default production dispatcher.

This table indexes evidence used for the main theorem, measured finite-slice claims, and Scheduleurm production-history bridge.

Table EC.5 Supplementary evidence index: external-policy and extension checks.

Gate family	Retained dashboard evidence	Boundary kept in manuscript
Policy-semantics baseline universe	Gavel-, Pollux-, Sia-, IADeep-, Salus-, fairness-, packing-, short-job, and delay-style policy rows evaluated on the same measured service cache.	Tests whether the candidate family omits useful finite scheduling semantics; it is not direct binary/full-stack superiority.
Gavel calibration and holdout rows	Adapter certificate, profile-aware calibration, resident-delay/JCT holdout, native same-workload simulator row, and same-host physical probe.	Keeps scalar service-unit equivalence separate from measured-cache policy comparison and scoped runtime evidence.
Named external runtime probes	Same-host runtime or adapter probes for Gavel, Pollux/AdaptDL, Sia, IADeep, and Salus, with paired workload and environment metadata when available.	Diagnostic implementation evidence only; original multi-node control-plane superiority is not claimed.
External action-union and algorithm upgrades	External policy-family actions, trajectory/action-semantics union rows, adaptive scalarized replay, state-dependent marginal lookup, global lookahead, and ETA reuse.	Shows the candidate family can absorb registered semantics; it is not production-default launch evidence.
Controlled migration-cost probes	Physical checkpoint, sync, and resume timing rows for controlled CNN, hybrid-RL, and CPU-surrogate payloads at 25%, 50%, and 75% progress.	Calibrates K_{mig} for benchmark migration actions only; ordinary running user jobs remain excluded.
Adjacent-domain scheduler bridge	Decima-style Spark-DAG simulator bridge, registered non-GPU scheduling diagnostics, and same-domain simulator notes.	Shows adjacent-domain coverage; it is not a GPU co-location full-stack comparison.
Learning and hidden-regime extensions	Active-bucket concentration, forced exploration, dwell/switching, detector-delay, optional live-hook records, and regime-detector fields.	Extension evidence; the current default live scheduler does not claim a complete online-learning theorem.
Non-future closure and future-state safety	Conservative all-state safety cover, probe/defer admission, zero theorem service for unknown states, and non-future closure rows excluding arbitrary future workloads.	Prevents overclaiming positive service or superiority for arbitrary future workloads, unregistered schedulers, or unmeasured hardware states.

This table keeps external scheduler and extension evidence visible while preserving the paper's scoped mathematical claim.

Code and Data Availability

The code, experiment artifacts, measured service-cache summaries, reviewer supplement manifest, and README files for this submission are provided in the supplementary Scheduleurm artifact package. The package contains the Scheduleurm source tree, theorem-facing replay modules, measured service-cache files, generated experiment summaries, and the Lean supplement headed by `ScheduleurmUpload.lean`. The reproduction script rebuilds the paper PDF, reruns the safe replay and certificate checks, executes the targeted tests, and refreshes the reproduction manifest. Live launches and external full-stack runtime probes are not executed by default; their recorded outputs are packaged with the corresponding environment notes. Raw scheduler history records that are outside the theorem-facing completed-active population are retained as operational telemetry and are not used as the claimed replication population. The Lean supplement package contains `ScheduleurmUpload.lean`, `lakefile.toml`, `lean-toolchain`, `build.log`, `theorem_crosswalk.md`, `manifest.json`, and checksums.